

STAT340 – Group Exercise 7

Classification

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Make sure you have installed and loaded the following packages and data before you start with today's exercises.

```
library(BIAS.data)
library(MASS)
library(mixlm)
library(tidyverse)
library(ggfortify)
data("poems")
data("poems.test")
```

Exercise 1 - Analysis of 'Deconstructed' Poems by PCA

Before we start with classification, let us have a look at the data set and explore it using PCA. The data frame `poems` contains letter frequencies collected from 22 poems written by either William Shakespeare (1564–1616), William Blake (1757–1827), and Thomas Stearns Eliot (1888–1965).

```
head(poems)
```

```
##      Author      a      b      c      d      e
## 1 Shakespeare 0.05110733 0.02214651 0.015332198 0.03407155 0.11754685
## 2 Shakespeare 0.05324459 0.01830283 0.014975042 0.03993344 0.11148087
## 3 Shakespeare 0.03754266 0.01706485 0.006825939 0.02901024 0.10750853
## 4 Shakespeare 0.04690117 0.01507538 0.011725293 0.03015075 0.12730318
## 5 Shakespeare 0.04974271 0.01372213 0.010291595 0.03087479 0.12521441
## 6 Shakespeare 0.04804270 0.01067616 0.014234875 0.03024911 0.09786477
##      f      g      h      i      j      k
## 1 0.01873935 0.020442930 0.05792164 0.04940375 0.000000000 0.003407155
## 2 0.01497504 0.009983361 0.05657238 0.04492512 0.000000000 0.004991681
## 3 0.02559727 0.015358362 0.06996587 0.05631399 0.000000000 0.005119454
## 4 0.02010050 0.010050251 0.04857621 0.05192630 0.000000000 0.001675042
## 5 0.02401372 0.010291595 0.06174957 0.05660377 0.000000000 0.005145798
## 6 0.01245552 0.035587189 0.06405694 0.06049822 0.001779359 0.008896797
##      l      m      n      o      p      q
## 1 0.03066440 0.01873935 0.04940375 0.04258944 0.005110733 0.000000000
## 2 0.04492512 0.01830283 0.04658902 0.04658902 0.009983361 0.000000000
## 3 0.04436860 0.02047782 0.03754266 0.07679181 0.010238908 0.000000000
## 4 0.04522613 0.01842546 0.04690117 0.04522613 0.006700168 0.005025126
```

```
## 5 0.03430532 0.01886792 0.04459691 0.05145798 0.017152659 0.001715266
## 6 0.04270463 0.01957295 0.05693950 0.06939502 0.012455516 0.000000000
##           r           s           t           u           v           w
## 1 0.05621806 0.05110733 0.10051107 0.02896082 0.003407155 0.02044293
## 2 0.03993344 0.06156406 0.06821963 0.03161398 0.004991681 0.03161398
## 3 0.04095563 0.06655290 0.07849829 0.02218430 0.006825939 0.02559727
## 4 0.05695142 0.05695142 0.08710218 0.02847571 0.011725293 0.03517588
## 5 0.04974271 0.04802744 0.10291595 0.02915952 0.005145798 0.01543739
## 6 0.04092527 0.04982206 0.06939502 0.02669039 0.007117438 0.01601423
##           x           y           z           !           ,           ;
## 1 0.000000000 0.02385009 0.000000000 0.000000000 0.02385009 0.000000000
## 2 0.001663894 0.03494176 0.001663894 0.001663894 0.01996672 0.000000000
## 3 0.000000000 0.01194539 0.000000000 0.000000000 0.01365188 0.001706485
## 4 0.001675042 0.02177554 0.001675042 0.000000000 0.01340034 0.003350084
## 5 0.000000000 0.01200686 0.000000000 0.000000000 0.01543739 0.005145798
## 6 0.000000000 0.02135231 0.000000000 0.001779359 0.02491103 0.003558719
##           space
## 1 0.1550256
## 2 0.1663894
## 3 0.1723549
## 4 0.1507538
## 5 0.1612350
## 6 0.1530249
```

The name of the poet is stored in the column `poems$Author`, and the last four columns contain frequencies for exclamation mark (!), comma (,), semicolon (;), and `space`, respectively.

There are 22 poems in total: 7 written by Blake, 7 written by Eliot, and 8 written by Shakespeare.

- a) Get acquainted with the data by visualizing letter frequencies in a bar plot. Compare the different authors. *(NB: Discuss the output, not the code)*

To compute a bar plot, we need to convert the data in a so-called 'long' table, since we need the data to be in three columns that contain information about the author, one for letters, and one for letter frequency.

```
poems.long <- poems %>%
  pivot_longer(-Author, names_to = "letter", values_to = "freq") %>%
  mutate(letter = as_factor(letter)) # convert 'letter' to factor to keep the order
poems.long
```

```
## # A tibble: 660 x 3
##   Author      letter  freq
##   <fct>      <fct>  <dbl>
## 1 Shakespeare a      0.0511
## 2 Shakespeare b      0.0221
## 3 Shakespeare c      0.0153
## 4 Shakespeare d      0.0341
## 5 Shakespeare e      0.118
## 6 Shakespeare f      0.0187
## 7 Shakespeare g      0.0204
## 8 Shakespeare h      0.0579
## 9 Shakespeare i      0.0494
## 10 Shakespeare j      0
## # i 650 more rows
```

After the data is converted, we can create a plot.

```
poems.long %>% ggplot(aes(x = letter, y = freq, fill = Author)) +
  geom_bar(stat = "identity") +      # adds the bar plot
  facet_grid(rows = vars(Author)) +  # this creates one plot per author
  xlab("Letter") +
  ylab("Letter frequency") +
  theme_light() +                   # cosmetics
  theme(legend.position = "none") +  # remove the legend
  scale_fill_viridis_d()            # use a colour scale other than default
```

- b) Conduct a principal component analysis using the `poem` data. What portion of the total variance in the data is captured jointly by the two first components? How many principal components are needed to explain 80 % of the variability?

```
pca <- prcomp(poems[, -1], scale=TRUE)
```

- c) Which variable (letter/character) has the largest influence (loading) on PC1 (have a look at `pca$rotation`)? Which variable is most influential on PC2?
- d) Create a biplot and comment (*NB: Discuss the output, not the code*). Do you think PC1 and PC2 are good variables for discriminating between the three authors? Justify your answer.

```
autoplot(pca,
  x = 1,          # plot PC1 on the x-axis
  y = 2,          # plot PC2 on the y-axis (play around with this)
  data = poems,   # you have to refer to the original data set for colour
  scale = 0,       # don't use the scaling implemented in autoplot()
  size = 4,        # size of points
  loadings = TRUE,
  loadings.label = TRUE,
  loadings.label.size = 4,
  loadings.label.repel = TRUE,
  loadings.label.colour = "grey70",
  loadings.colour = "grey70",
  colour = "Author") +
  scale_colour_viridis_d() +
  theme_minimal()
```

Exercise 2 - Predicting Authorship by LDA

Six additional poems, two from each author, were collected as test samples, and their letter frequencies were computed. These poems are stored in the file `poemsTest.RData`. The six new poems were:

##	Author	a	b	c	d	e	
## 1	Shakespeare	0.04448399	0.02313167	0.01779359	0.02669039	0.11565836	
## 2	Shakespeare	0.06040268	0.01342282	0.01342282	0.02516779	0.09563758	
## 3	Blake	0.05021521	0.01291248	0.01291248	0.03443329	0.11621234	
## 4	Blake	0.06611570	0.01652893	0.01446281	0.04545455	0.09917355	
## 5	Eliot	0.06306306	0.02402402	0.02602603	0.03203203	0.11111111	
## 6	Eliot	0.06510851	0.01335559	0.02003339	0.05175292	0.12353923	
##		f	g	h	i	j	k

```
## 1 0.023131673 0.02135231 0.06049822 0.03202847 0.000000000 0.001779359
## 2 0.023489933 0.01510067 0.06375839 0.04362416 0.000000000 0.008389262
## 3 0.007173601 0.02008608 0.06169297 0.03730273 0.002869440 0.011477762
## 4 0.018595041 0.01033058 0.04752066 0.05578512 0.002066116 0.002066116
## 5 0.019019019 0.01701702 0.04204204 0.05405405 0.002002002 0.013013013
## 6 0.020033389 0.01001669 0.05843072 0.04674457 0.000000000 0.006677796
##          l          m          n          o          p          q
## 1 0.03202847 0.007117438 0.04626335 0.05693950 0.007117438 0.001779359
## 2 0.03187919 0.016778523 0.04362416 0.07382550 0.010067114 0.000000000
## 3 0.04017217 0.014347202 0.06456241 0.06169297 0.011477762 0.000000000
## 4 0.03099174 0.008264463 0.06818182 0.04338843 0.010330579 0.000000000
## 5 0.03803804 0.020020020 0.05705706 0.04904905 0.015015015 0.000000000
## 6 0.03005008 0.016694491 0.06510851 0.05342237 0.008347245 0.001669449
##          r          s          t          u          v          w
## 1 0.02669039 0.06939502 0.09786477 0.05160142 0.016014235 0.017793594
## 2 0.04530201 0.05201342 0.10402685 0.02852349 0.015100671 0.006711409
## 3 0.06456241 0.06456241 0.06025825 0.02725968 0.001434720 0.014347202
## 4 0.06611570 0.04752066 0.05371901 0.01859504 0.008264463 0.012396694
## 5 0.05205205 0.07607608 0.06606607 0.03403403 0.003003003 0.015015015
## 6 0.05342237 0.06176962 0.06510851 0.02170284 0.008347245 0.023372287
##          x          y          z          !          ,          ;
## 1 0.001779359 0.024911032 0.000000000 0.000000000 0.021352313 0.000000000
## 2 0.000000000 0.021812081 0.000000000 0.003355705 0.018456376 0.001677852
## 3 0.000000000 0.021520803 0.000000000 0.000000000 0.022955524 0.004304161
## 4 0.000000000 0.016528926 0.000000000 0.002066116 0.016528926 0.000000000
## 5 0.003003003 0.008008008 0.003003003 0.001001001 0.005005005 0.007007007
## 6 0.000000000 0.010016694 0.000000000 0.000000000 0.005008347 0.000000000
##          space
## 1 0.1548043
## 2 0.1644295
## 3 0.1592539
## 4 0.2190083
## 5 0.1441441
## 6 0.1602671
```

In this exercise we will use the data from the previous exercise (=training data), and try to classify the new poems (=test data).

- a) Compute the PC1- and PC2-scores of the poems in `poems.test` using the PCA from Exercise 1b).
Hint: The `predict()` function is quite useful for such purposes.

```
scores.test <- predict(pca, poems.test)
scores.test <- cbind(Author = poems.test$Author, as_tibble(scores.test))
scores.test
```

```
##          Author      PC1      PC2      PC3      PC4      PC5
## 1 Shakespeare  3.9136323  1.3319443 -1.1169660 -1.7389198 -2.32377638
## 2 Shakespeare  1.2103595  0.8718919  1.4035316 -1.4831095 -1.37059168
## 3      Blake   0.3442297 -0.8672537  1.9535695 -1.6928408 -0.91702562
## 4      Blake  -0.9636156 -1.9237165 -0.2112698 -0.2821465 -0.38333485
## 5      Eliot   3.0454523 -1.3419746 -1.6560653 -1.8836301  2.52978944
## 6      Eliot   1.4799615 -0.7610281 -0.5837241 -0.2428388 -0.01116067
##          PC6      PC7      PC8      PC9      PC10      PC11
```

```
## 1 -0.9397343 -0.00599067 -1.7846588 0.10208251 1.2352745 1.13913685
## 2 -1.1495221 -1.96209935 0.7949635 0.55409984 0.7704929 -0.23480134
## 3 1.3826043 -0.98275305 -0.1448373 -0.63678061 -0.4290403 1.30888775
## 4 -0.6413104 -0.81644363 -1.0130836 0.32640032 -0.5669068 -1.05881691
## 5 2.5002166 1.17765519 -0.4908587 -3.61870220 0.7830911 -1.19853250
## 6 0.9840082 -1.21568818 0.6548695 -0.09575747 0.9862488 -0.02946278
##      PC12      PC13      PC14      PC15      PC16      PC17
## 1 0.4091209 1.2356087 -0.2016212 1.5615912 2.39204801 0.006314123
## 2 -0.5084311 1.2270572 1.3450224 0.8736042 1.53476996 -0.496565396
## 3 0.2654943 -0.2313682 -0.3462967 -0.1791460 -0.23224049 -0.733424107
## 4 -1.7870994 -1.0446252 -1.0192751 0.3371612 -0.02075176 1.450085633
## 5 0.1534951 1.6249199 0.4313415 -1.7051011 -1.39612831 -1.071581614
## 6 -0.5017367 -0.9360946 -0.9680239 1.3120686 0.77474374 0.978942809
##      PC18      PC19      PC20      PC21      PC22
## 1 -1.47471763 -1.1302691 0.78916096 -1.9026131 -0.93676072
## 2 -0.20637255 -0.7571421 0.33141908 -0.9983007 1.13014308
## 3 0.36100292 -1.6395639 -1.88463268 0.2649253 -0.00584988
## 4 0.41879891 1.2680529 0.21975369 -0.3832447 0.09770024
## 5 -0.26276794 -0.2117775 -0.02190933 0.3644742 1.02446912
## 6 0.04921075 0.5126721 -0.39308447 0.8671660 -0.85642192
```

- b) Conduct a linear discriminant analysis (LDA) using PC1 and PC2 from exercise 1b) as predictors to classify the six 'test' poems to either Shakespeare, Blake or Eliot.

```
# First, we need to make a new data frame containing the information
# about the author and the first two principal components.
scores.poems <- cbind(Author = poems$Author, as_tibble(pca$x))

# Now, let's do the LDA
lda.poems <- lda(Author ~ PC1 + PC2,
                 data = scores.poems)

# Then, predict the authorship of the 'test' poems
lda.pred <- predict(lda.poems, newdata = scores.test)
```

- c) The posterior probabilities are stored in `lda.pred$posterior`. Display the values using `round(lda.pred$posterior, digits = 2)`. What is the predicted authorship of the different poems?
- d) Write up a so-called confusion matrix for the test classification with true author as rows and classified authors as columns. Use the `confusion()` function from the package `mixlm`. What poems from which author are most difficult to predict? Compute the apparent error rate (APER) and the accuracy.

```
lda.pred <- predict(lda.poems, newdata = scores.test)
mixlm::confusion(poems.test$Author, # true classes
                 lda.pred$class)    # predicted classes
```

```
##      True
## Predicted Blake Eliot Shakespeare
## Blake      0      0      0
## Eliot       2      1      0
## Shakespeare 0      1      2
## Total      2      2      2
## Correct    0      1      2
```

```
##
## Proportions correct
##      Blake      Eliot Shakespeare
##      0.0      0.5      1.0
##
## N correct/N total = 3/6 = 0.5
```

- e) How would you think one can improve the LDA? By adding more ‘test’ data (more observations in `poems.test`) or more ‘training’ data (data that the LDA is fit on, i.e. `poems`)?